Uniformly smooth Banach spaces fail Property P

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Source Question

Mandal–Manna–Paul–Sain introduce an approximate-preserver version of the Blanco–Koldobsky–Turnsek theorem [1]. In their notation, $x \perp_B^{\varepsilon} y$ means that

$$\|x + \lambda y\|^2 \geq \|x\|^2 - 2\varepsilon \|x\| \|\lambda y\| \qquad (\lambda \in \mathbb{R}).$$

An operator $T \in \mathcal{L}(X)$ preserves ε-orthogonality at x if $x \perp_B y$ implies $Tx \perp_B^{\varepsilon} Ty$ for all $y \in X$. The paper asks whether, for a given Banach space X, there exists $\varepsilon \in (0, 1)$ such that every operator preserving ε-orthogonality at all points of X is a scalar multiple of an isometry. This property is called Property P.

In view of the Blanco-Koldobsky-Turnšek characterization of isometries, it is natural to consider the following more general query:

Given a Banach space $\mathbb{X}$, does there exist an $\epsilon \in (0,1)$, such that any $T \in \mathbb{L}(\mathbb{X})$ preserving ϵ-orthogonality is an isometry multiplied by a constant?

It is clear that an affirmative answer to the above query will directly lead to a proper refinement of the Blanco-Koldobsky-Turnšek theorem, which is the main motivation behind this study. In order to address the above query, it is convenient to introduce the following definition:

Definition 1.2. Let $\mathbb{X}$ be a Banach space. We say that $\mathbb{X}$ has the *Property P* if there exists an $\epsilon \in (0,1)$ such that the following two conditions are equivalent:
(i) $T \in \mathbb{L}(\mathbb{X})$ preserves ϵ-orthogonality at each $x \in \mathbb{X}$.
(ii) $T \in \mathbb{L}(\mathbb{X})$ is a scalar multiple of an isometry.

Clearly, whenever a Banach space $\mathbb{X}$ possesses the *Property P*, a refinement of the Blanco-Koldobsky-Turnšek theorem is possible for $\mathbb{X}$. On the other hand, if $\mathbb{X}$ does not possess the *Property P*, it follows that the Blanco-Koldobsky-Turnšek theorem can not be improved by considering approximate preservation of orthogonality. Our main aim is to produce concrete examples of both types of Banach spaces by considering certain geometric properties of the concerned space. We first investigate some geometric and analytic properties of finite-dimensional polyhedral Banach spaces, including the smoothness and the number of extreme supporting functionals of each points, related to such preservation. Additionally, we analyze the linear operators satisfying such preservation of orthogonality, in connection with the geometry of the concerned spaces. As an application of this study, we characterize the isometries on some Banach spaces, thus obtaining refinements of the Blanco-Koldobsky-Turnšek theorem on these spaces. Finally, we establish that a two-dimensional Banach space, whose unit sphere is given by a regular $2n$-gon $(n > 2)$, and the sequence space ℓ_∞^n $(n > 2)$ do satisfy the *Property P*. On the other hand, we prove that the sequence spaces ℓ_p $(1 \leq p < \infty)$ do not possess this property.

We end this section with the following characterization of the approximate Birkhoff-James orthogonality, which will be used repeatedly.

Theorem 1.3. [4, Th. 2.4] *Let $\mathbb{X}$ be a Banach space and let $x, y \in \mathbb{X}$. Let $\epsilon \in [0,1)$. Then x is ϵ-approximate Birkhoff-James orthogonal to y if and only if there exists $f \in J(x)$ such that $|f(y)| \leq \epsilon \|y\|$.*

2. Main Results

We begin this section with the following proposition.

Proposition 2.1. *Let $\mathbb{X}$ be a finite-dimensional polyhedral Banach space. For each $f \in Ext\ B_{\mathbb{X}^*}$, consider the set $\mathcal{A}_f = \{x \in \mathbb{X} : J(x) = \{f\}\}$. Then the following results hold true:*

The source proves that regular $2n$-gon planes with $n > 2$ and ℓ_∞^n, $n > 2$, have Property P, while ℓ_p and ℓ_p^n, $1 \leq p < \infty$, do not. The result below gives a broad negative theorem on the smooth side of the problem.

Background and Definitions

For a nonzero $x \in X$, let
$$J(x) = \{f \in S_{X^*} : f(x) = \|x\|\}$$
be the set of unit supporting functionals at x. The source cites the Chmielinski–Stypula–Wojcik characterization [2, Theorem 2.4]: for $\varepsilon \in [0, 1)$,

$$x \perp_B^{\varepsilon} y \quad \Longleftrightarrow \quad \text{there is } f \in J(x) \text{ with } |f(y)| \leq \varepsilon\|y\|.$$

In particular, in a smooth real Banach space, writing $j(x)$ for the unique member of $J(x)$, exact Birkhoff–James orthogonality is equivalent to $j(x)(y) = 0$.

We use the standard fact that if X is uniformly smooth, then the normalized duality mapping is uniformly norm-to-norm continuous on bounded subsets [3, Chapter I, Section 4]. In the source's unit-support normalization, this implies that $x \mapsto j(x)$ is uniformly continuous on every annulus $\{x : a \leq \|x\| \leq b\}$ with $0 < a < b < \infty$.

Proof Intuition

The obstruction is local but uniform. In a uniformly smooth space, a small operator-norm perturbation of the identity moves the supporting functional $j(x)$ by a uniformly small amount for all $x \in S_X$. Therefore exact orthogonality $j(x)(y) = 0$ remains approximate orthogonality after applying the perturbation, provided the perturbation is small enough.

The perturbation can be chosen so that it is definitely not a similarity: take a rank-one shear $T_\delta z = z + \delta\phi(z)v$, where $v \in \ker\phi$. It fixes the hyperplane $\ker\phi$ pointwise but shifts every affine line parallel to v outside that hyperplane. A scalar multiple of an isometry cannot do this, because it would make the convex function $t \mapsto \|w + tv\|$ periodic.

Theorem 1. *Let X be a real uniformly smooth Banach space with $\dim X > 1$. Then X does not have Property P.*

Proof. It is enough to prove that for every $0 < \varepsilon < 1$ there is an operator on X that preserves ε-orthogonality at all points but is not a scalar multiple of an isometry.

Choose $v \in S_X$. Since $\dim X > 1$, there is a norm-one functional $\phi \in X^*$ such that $\phi(v) = 0$. Put
$$Az = \phi(z)v, \qquad T_\delta = I + \delta A.$$
Then $\|A\| = 1$, and if $0 < \delta < 1$, T_δ is invertible by the Neumann series.

First we check that T_δ is not a scalar multiple of an isometry when $\delta \neq 0$. Let $H = \ker\phi$. The operator T_δ fixes H pointwise. If $T_\delta = cU$, where U is an isometry, then for any nonzero $h \in H$,

$$\|h\| = \|T_\delta h\| = |c|\|Uh\| = |c|\|h\|,$$

so $|c| = 1$. Hence T_δ itself would be norm preserving. Choose $w \in X$ with $\phi(w) = 1$. Since $v \in H$,

$$T_\delta(w + tv) = w + (t + \delta)v \qquad (t \in \mathbb{R}).$$

Norm preservation would imply that
$$q(t) = \|w + tv\|$$
is δ-periodic. But q is convex, and a finite convex periodic function on $\mathbb{R}$ is constant, while $q(t) \geq |t| - \|w\| \to \infty$. This is impossible.

Now fix $0 < \varepsilon < 1$. By uniform continuity of j on $\{z : 1/2 \le \|z\| \le 3/2\}$, choose $0 < \delta < 1/2$ so small that

$$\|j(T_\delta x) - j(x)\| \le \varepsilon/2 \qquad (x \in S_X)$$

and also

$$\frac{\delta}{1-\delta} \le \varepsilon/2.$$

This is possible because $\|T_\delta x - x\| \le \delta$ and $\|T_\delta x\| \in [1-\delta, 1+\delta]$ for $x \in S_X$.

Let $x, y \in S_X$ and suppose $x \perp_B y$. Smoothness gives $j(x)(y) = 0$. Since $T_\delta y = y + \delta\phi(y)v$, we have

$$\begin{aligned} |j(T_\delta x)(T_\delta y)| &\le |(j(T_\delta x) - j(x))(T_\delta y)| + |j(x)(T_\delta y)| \\ &\le (\varepsilon/2)\|T_\delta y\| + \delta|\phi(y)|\,|j(x)(v)| \\ &\le (\varepsilon/2)\|T_\delta y\| + \delta. \end{aligned}$$

Because $\|T_\delta y\| \ge 1 - \delta$, the second choice of δ gives $\delta \le (\varepsilon/2)\|T_\delta y\|$. Therefore

$$|j(T_\delta x)(T_\delta y)| \le \varepsilon\|T_\delta y\|.$$

By the Chmielinski–Stypula–Wojcik characterization, this means

$$T_\delta x \perp_B^\varepsilon T_\delta y.$$

The cases $x = 0$ or $y = 0$ are trivial, and homogeneity in both variables extends the conclusion from unit vectors to arbitrary $x, y \in X$. Thus T_δ preserves ε-orthogonality at every point of X, while it is not a scalar multiple of an isometry. Since $\varepsilon \in (0,1)$ was arbitrary, no ε can witness Property P for X. □

Corollary 1. *Every real Hilbert space of dimension greater than one, and more generally every real ℓ_p-type or L_p-type uniformly smooth Banach space with dimension greater than one and $1 < p < \infty$, fails Property P.*

Verification and Novelty Check

The proof uses only three external ingredients: the source definition of Property P, the Chmielinski–Stypula–Wojcik support-functional characterization of ε-Birkhoff–James orthogonality, and the standard uniform continuity of the duality map in uniformly smooth spaces.

The argument is insensitive to dimension, including finite-dimensional smooth spaces. It does not contradict the source's positive examples: regular $2n$-gon planes and ℓ_∞^n are polyhedral and nonsmooth.

Bounded duplicate/literature check performed before promotion:

- local run indexes were searched for arXiv:2501.10719, Property P, approximate preservation of Birkhoff–James orthogonality, and uniformly smooth variants;
- web search located the published Colloquium Mathematicum entry for the source paper and the later Lebesgue–Bochner continuation arXiv:2601.02786;
- no exact statement that all real uniformly smooth spaces fail Property P was found in the bounded search.

Human-review recommendation: check the normalization of the duality map against the source's $J(x)$ convention and verify that the final homogeneity step matches the source's ε-orthogonality definition. These are the only places where conventions can hide small errors; the shear mechanism itself is elementary.

References

[1] K. Mandal, J. Manna, K. Paul, and D. Sain, *On approximate preservation of orthogonality and its application to isometries*, arXiv:2501.10719; Colloquium Mathematicum 179 (2025), 189–211.

[2] J. Chmielinski, K. Stypula, and P. Wojcik, *Approximate Birkhoff orthogonality*, J. Math. Anal. Appl. 461 (2018), 308–324.

[3] I. Cioranescu, *Geometry of Banach Spaces, Duality Mappings and Nonlinear Problems*, Kluwer Academic Publishers, 1990.

[4] Mohit and R. Jain, *Approximate Birkhoff-James orthogonality preserver on Lebesgue-Bochner spaces*, arXiv:2601.02786, 2026.